

CO1-AT1-CODING CHALLENGE

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COURSE CODE: CSA0911

QUESTION-10:

CODE:

```
public class Device {  
    void on() {  
        System.out.println("Device ON");  
    }  
    void off() {  
        System.out.println("Device OFF");  
    }  
    void powerConsumption() {}  
    public static void main(String[] args) {  
        Device[] devices = {  
            new Light(),  
            new Fan(),  
            new AirConditioner();  
        }  
        for (Device d : devices) {  
            d.on();  
            d.powerConsumption();  
            d.off();  
            System.out.println();  
        }  
    }  
}  
  
class Light extends Device {  
  
    @Override  
    void powerConsumption() {
```

```
        System.out.println("Light = 20W");}}}
```

```
class Fan extends Device {
```

```
    @Override
```

```
    void powerConsumption() {
```

```
        System.out.println("Fan = 75W"); }}
```

```
class AirConditioner extends Device {
```

```
    @Override
```

```
    void powerConsumption() {
```

```
        System.out.println("AC = 1500W"); }}
```

OUTPUT:

A screenshot of a Java IDE interface. The left pane shows the source code for 'Device.java'. The code defines a 'Device' class with 'on()', 'off()', and 'powerConsumption()' methods, and a 'main' method. The right pane shows the 'Output' window with the results of running the program. The output shows three separate runs: 1) 'Device ON', 'Light = 20W', 'Device OFF'; 2) 'Device ON', 'Fan = 75W', 'Device OFF'; 3) 'Device ON', 'AC = 1500W', 'Device OFF'. The IDE has a dark theme and includes icons for running, debugging, and sharing.